

Schematics Only for A3K5GE1

NAME A3K5GE1_V01

Main Functions Introduction

- 1) Support::

Debug UART, Debug SWD (4Pin)
- 2) Support::

SPI
- 3) Support::

UART
- 4) Support::

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- 5) Support::

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- 6) Support::

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- 7) Support::

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- 8) Support::

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- 9) Support::

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- 10) Support::

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- 11) Support::

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- 12) Support::

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- 13) Support::

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- 14) Support::

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- 15) Support::

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 昇科源科技 SKYAUTO TECH				
Project:	A3K5GE1			
File:	00.Cover Page			
Date:	Monday, April 21, 2025			Rev: V0_1
Designed by:	**	Reviewed by:	**	Sheet: 1 of 21

[illegible]

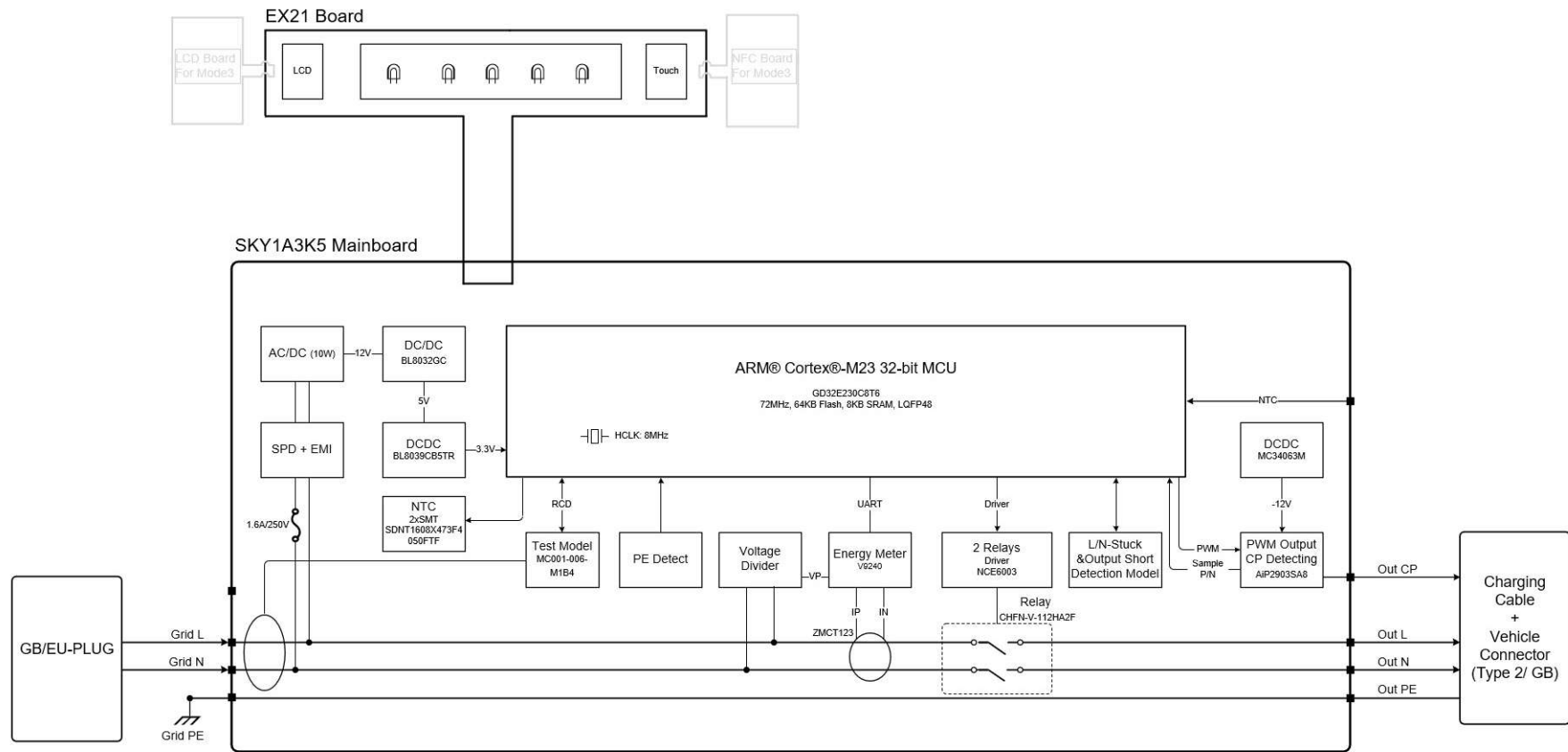
Item	Value	Description	PCB Footprint	Reference	Quantity	Option
1	1	Resistor	0603	R1	1	
2	1	Capacitor	0603	C1	1	
3	1	IC	0603	U1	1	
4	1	Diode	0603	D1	1	
5	1	Inductor	0603	L1	1	
6	1	Transistor	0603	Q1	1	
7	1	Relay	0603	Y1	1	
8	1	Switch	0603	S1	1	
9	1	Connector	0603	X1	1	
10	1	LED	0603	U2	1	

Option

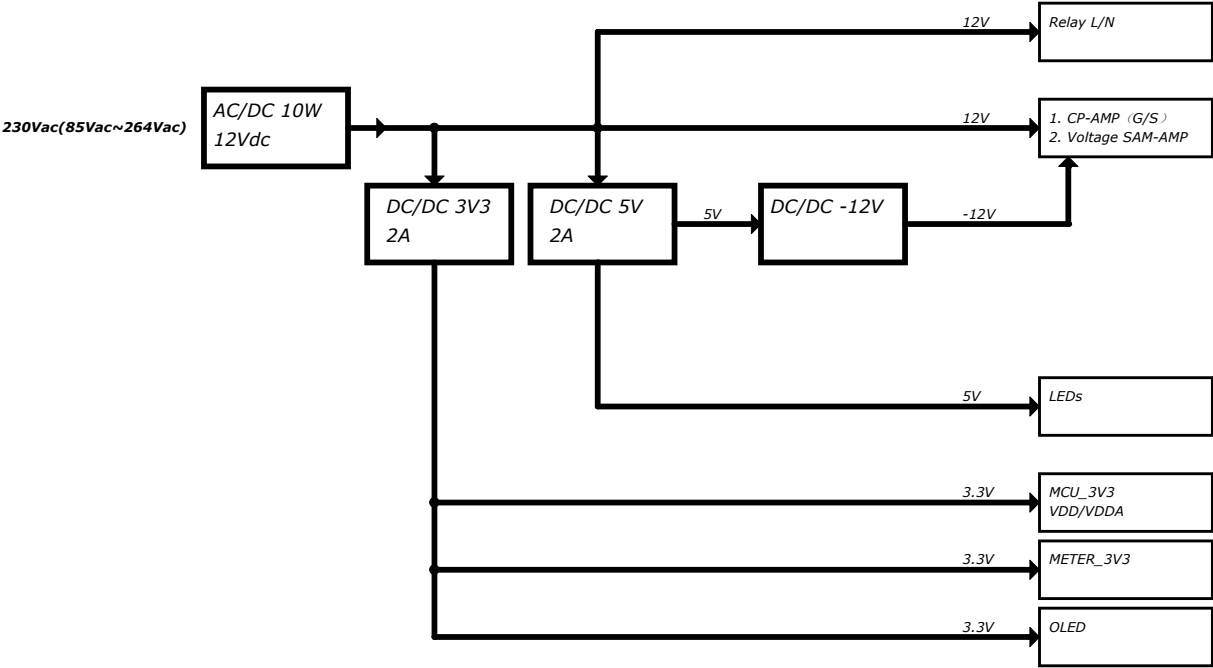
 昇科源科技 SKYAUTOTECH			
Project:	A3K5GE1		
File:	01.Index and Notes		
Date:	Monday, April 21, 2025		Rev: **
Designed by:	**	Reviewed by:	**
		Sheet:	2 of 21

Revision History

Version	Date	By	Change Dscription	Approved
V01	20250418	Dong	1.增加C1312 暂时不贴 2. 去掉FB802	
V01	20250421	Dong	J1801 引脚顺序更换 增加 R1806 R1807 R1808 R1809 R1810 R1811 R1812 R1813 R1814 增加Q1802 Q1803 R1312 R1333换成22ohm	



Power Tree



0501A
GD32E230C8T6
LQFP48-7x7mm

PA0/ USART0_CTS, USART1_CTS, CMP_OUT, I2C1_SCL/ ADC_IN0, CMP_IN6, RTC_TAMP1, WKUP0	10	<<MCU_PA0_SampleCP_P-ADC_IN0
PA1/ USART0_RTS, USART1_RTS, I2C1_SDA, EVENTOUT, TIMER14_CH0_ON/ ADC_IN1, CMP_IN7	11	<<MCU_PA1_SampleCP_N-ADC_IN1
PA2/ USART0_TX, USART1_TX, TIMER14_CH0/ ADC_IN2, CMP_IN7	12	<<MCU_PA2_DEBUG-USART1_TX-OutputShortDetectionH
PA3/ USART0_RX, USART1_RX, TIMER14_CH1/ ADC_IN3	13	<<MCU_PA3_OutputShortDetectionH
PA4/ SPI0_NSS, I2S0_WS, USART0_CK, USART1_CK, TIMER13_CH0, SPI1_NSS/ ADC_IN4, CMP_IN4	14	<<MCU_PA4_StuckL_Detection_ADC_IN4
PA5/ SPI0_SCK, I2S0_CK/ ADC_IN5, CMP_IN5	15	<<MCU_PA5_StuckM_Detection_ADC_IN5
PA6/ SPI0_MISO, I2S0_MCK, TIMER2_CH0, TIMER0_BRKIN, TIMER15_CH0, EVENTOUT, CMP_OUT/ ADC_IN6	16	<<MCU_PA6_NTC1_Detection-ADC_IN6
PA7/ SPI0_MOSI, I2S0_SD, TIMER2_CH1, TIMER13_CH0, TIMER0_CH0_ON, TIMER16_CH0, EVENTOUT/ ADC_IN7	17	<<MCU_PA7_NTC2_Detection-ADC_IN7
PA8/ USART0_CK, TIMER0_CH0, CK_OUT, USART1_TX, EVENTOUT	29	<<MCU_PA8_PWM-0_CP_TIMER0_CH0
PA9/ USART0_TX, TIMER0_CH1, TIMER14_BRKIN, I2C0_SCL, CK_OUT	30	<<MCU_PA9_METER_USART0_TX
PA10/ USART0_RX, TIMER0_CH2, TIMER16_BRKIN, I2C0_SDA	31	<<MCU_PA10_METER_USART0_RX
PA11/ USART0_CTS, TIMER0_CH3, CMP_OUT, EVENTOUT, SPI1_IOS, I2C0_SNSA, I2C1_SCL	32	<<MCU_PA11_PWM-0_LED_TIMER0_CH3
PA12/ USART0_RTS, TIMER0_ETI, EVENTOUT, SPI1_IOS(5), I2C0_TXFRAME, I2C1_SDA(5)	33	<<MCU_PA12_PMSV_EN
PA15/ SPI0_NSS, I2S0_WS, USART0_RX, USART1_RX, SPI1_NSS, EVENTOUT	38	<<MCU_PA15_DEBUG-USART1_RX

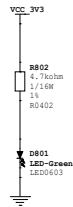
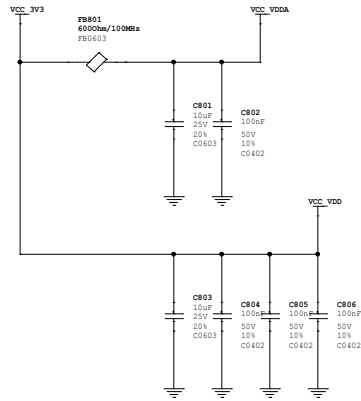
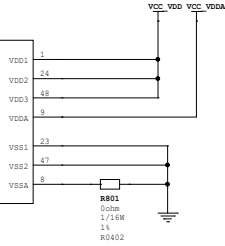
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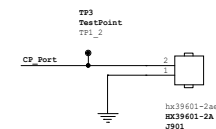
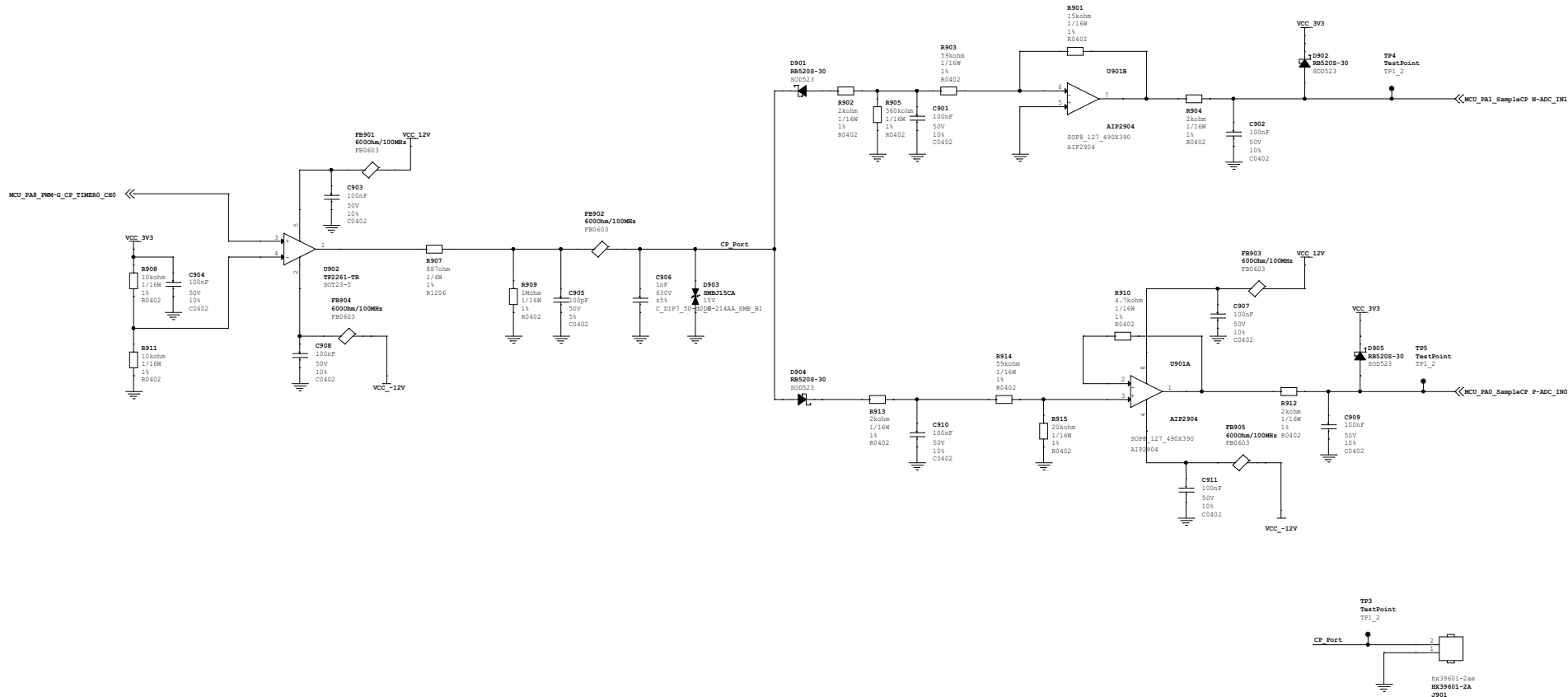
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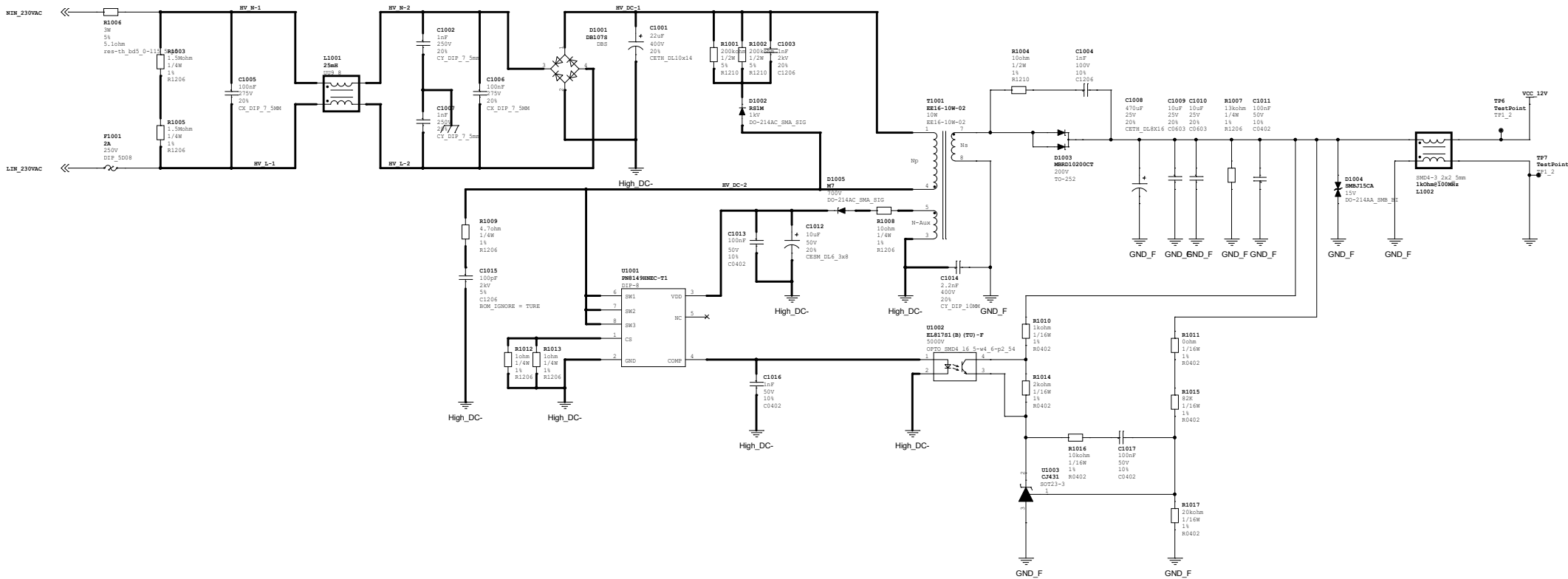
FB0/ TIMER2_CH2, TIMER0_CH1_ON, USART1_RX, EVENTOUT/ ADC_IN8	18	⋈⋈MCU_FB0_RTC3_Detection-ADC_IN8
FB1/ TIMER2_CH3, TIMER13_CH0, TIMER0_CH2_ON, SPI1_SCK/ ADC_IN9	19	⋈⋈MCU_FB1_InputPE_Detection-ADC_IN9
FB2/ TIMER2_ETI	20	⋈⋈MCU_FB2_Touch_Detection
FB10/ I2C0_SCL, I2C1_SCL, SPI1_I2C, SPI1_SCK	21	⋈⋈MCU_FB10_Relay_Driver L_EN
FB11/ I2C0_SDA, I2C1_SDA, EVENTOUT, SPI1_I2C	22	⋈⋈MCU_FB11_Relay_Driver W_EN
FB12/ SPI0_NSS, SPI1_NSS, TIMER0_BRKIN, I2C1_SMBA, EVENTOUT	23	⋈⋈MCU_FB12_LCD_SPI0_NSS
FB13/ SPI0_SCK, SPI1_SCK, TIMER0_CH0_ON, I2C1_TXFRAME, I2C1_SCL	24	⋈⋈MCU_FB13_LCD_SPI1_SCK
FB14/ SPI0_MISO, SPI1_MISO, TIMER0_CH1_ON, TIMER14_CH0, I2C1_SDA	27	⋈⋈MCU_FB14_LCD_SPI2_MISO
FB15/ SPI0_MOSI, SPI1_MOSI, TIMER0_CH2_ON, TIMER14_CH1, RTC_REFIN, WKUP6	28	⋈⋈MCU_FB15_LCD_SPI3_MOSI
FB3/ SPI0_SCK, I2S0_CK, EVENTOUT	39	⋈⋈MCU_FB3_RCD-Trip_Detection
FB4/ SPI0_MISO, I2S0_MCK, TIMER2_CH0, EVENTOUT, I2C0_TXFRAME, TIMER16_BRKIN	40	⋈⋈MCU_FB4_RCD-Zero_Cal
FB5/ SPI0_MOSI,I2S0_SD, I2C0_SMBA, TIMER15_BRKIN, TIMER2_CH1/ WKUP5	41	⋈⋈MCU_FB5_RCD-RMS_Cal
FB6/ I2C0_SCL, USART0_TX, TIMER15_CH0_ON	42	⋈⋈MCU_FB6_LCD_VCC3V3_EN
FB7/ I2C0_SDA, USART0_RX,TIMER16_CH0_ON	43	⋈⋈MCU_FB7_RCD-RMS_Test
FB8/ I2C0_SCL, TIMER15_CH0	45	⋈⋈MCU_FB8_Meter_INT
FB9/ I2C0_SDA, IFRP_OUT, TIMER16_CH0,EVENTOUT, I2S0_MCK, SPI1_NSS	46	⋈⋈MCU_FB9_OutputShortDET

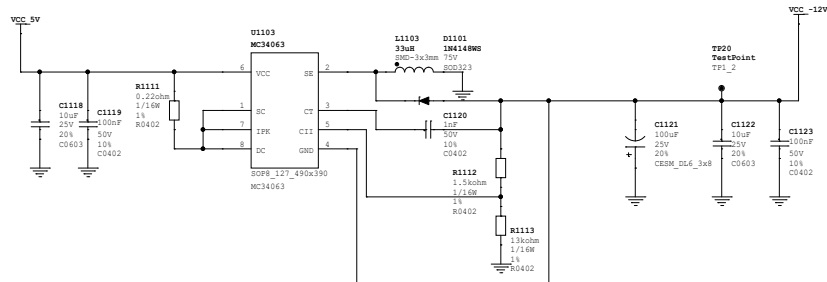
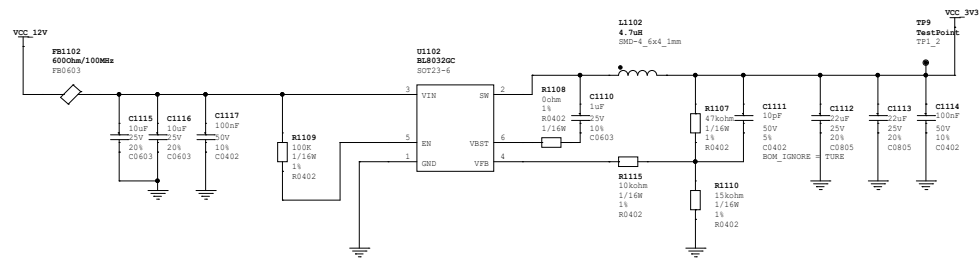
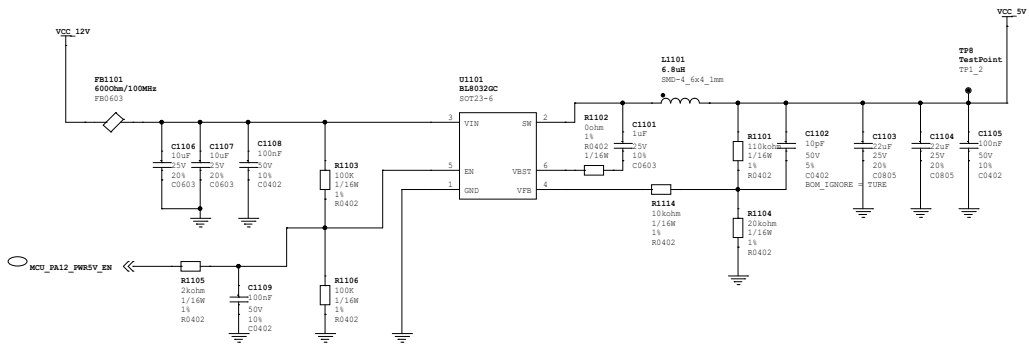
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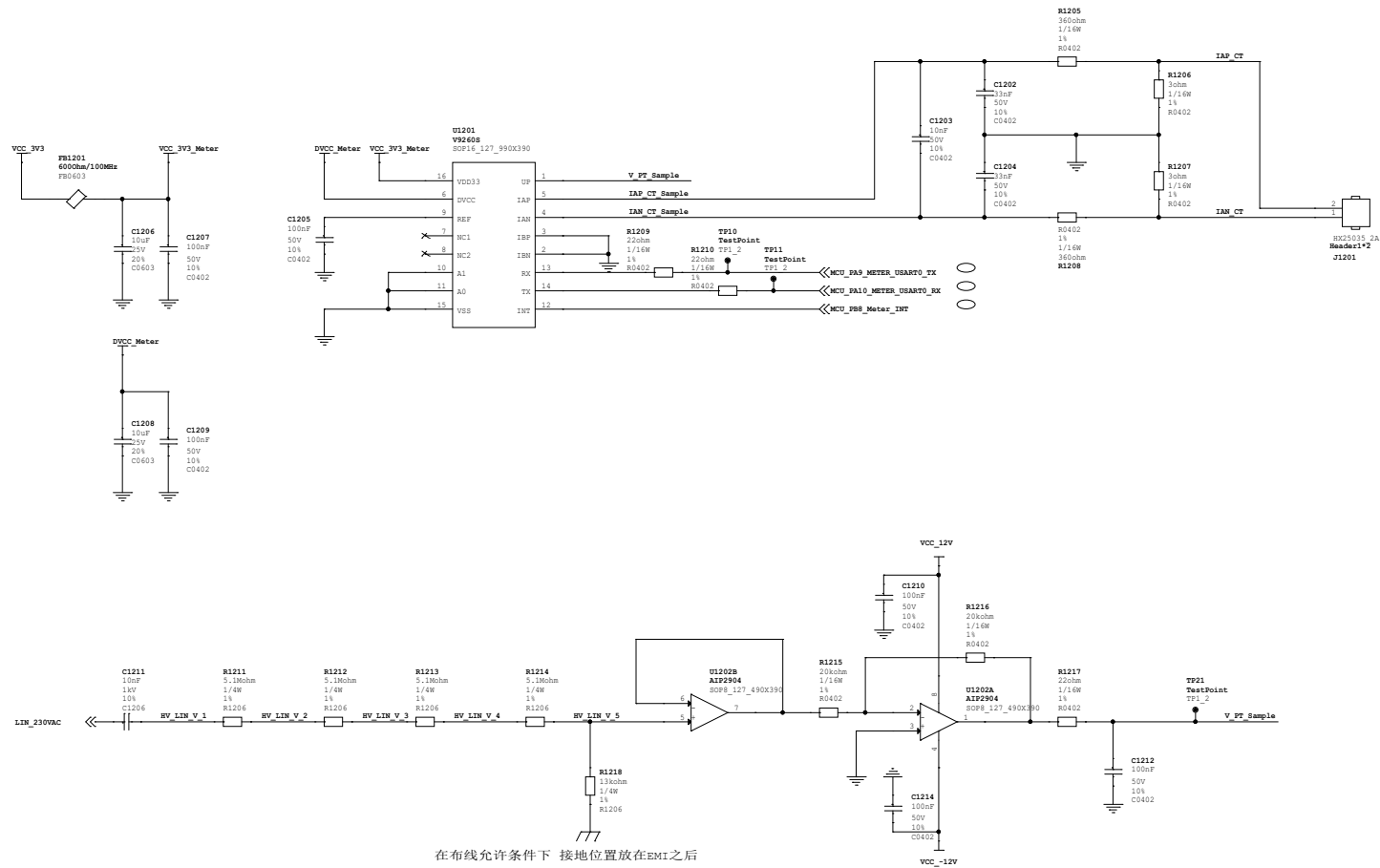
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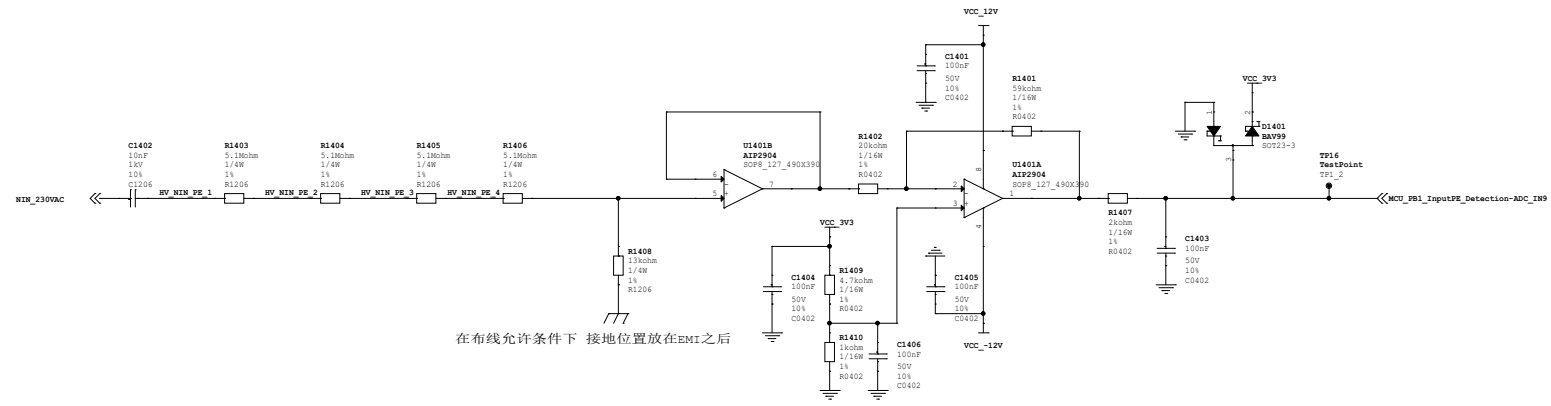


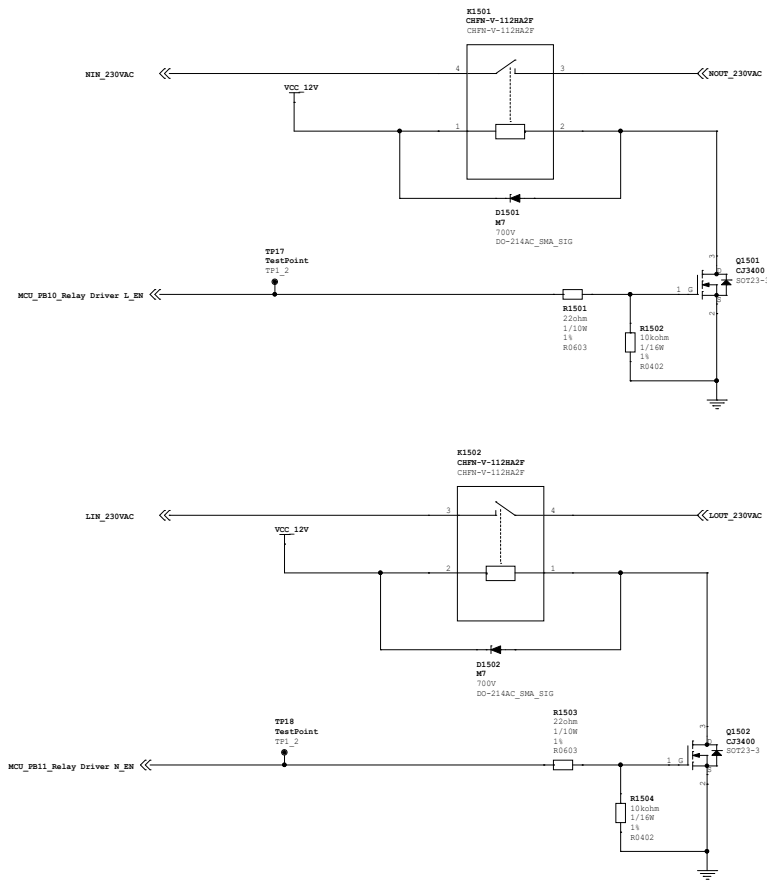


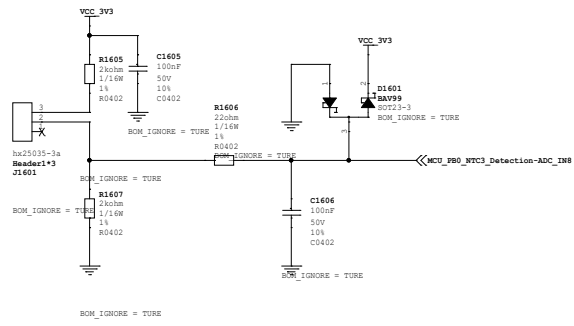
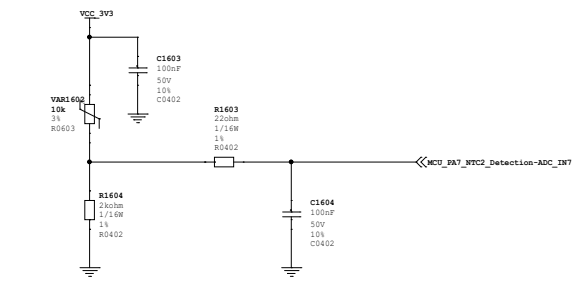
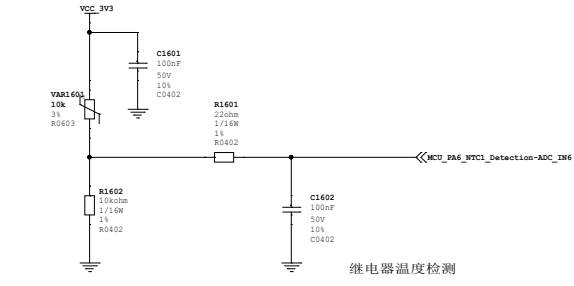


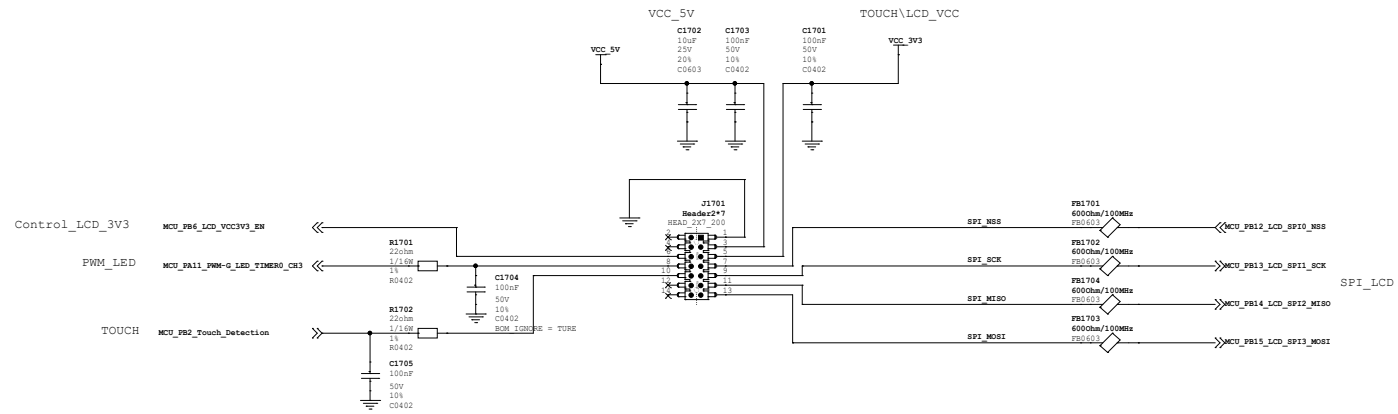


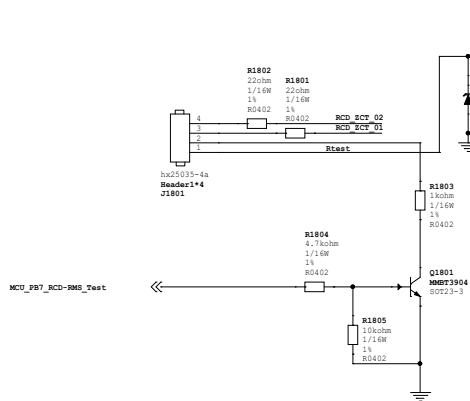
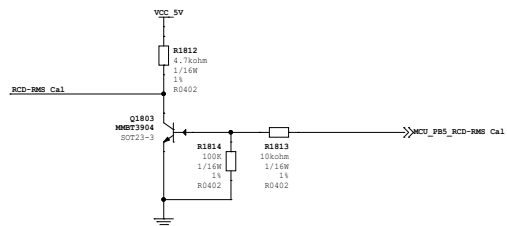
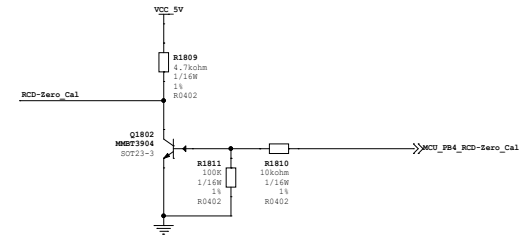
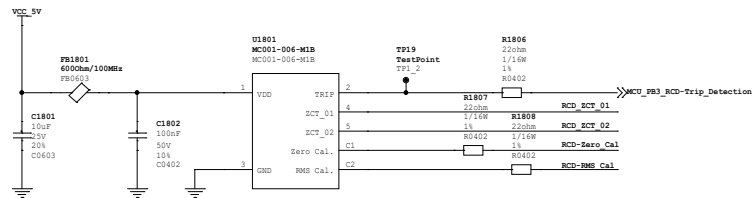












* Rtest根据自检过程需要模拟的直流剩余电流大小而定，互感器内部的自检线圈匝数是2匝。
举个例子，如果要产生直流10mA的自检电流，自检线圈接入的是5V电压，那么Rtest=5V/(10mA/2)=1k。

